

Transdimensional Seismic Tomography Using Reversible Jump MCMC with Voronoi Parameterization

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In this project, we apply reversible jump Markov Chain Monte Carlo (rj-MCMC) with a Voronoi cell parameterization to a simplified geophysical inverse problem inspired by seismic tomography. We construct a surrogate model of the subsurface, represented as a square domain containing three anomalies with distinct Rayleigh wave velocities, embedded in a uniform background.

Using a forward model, we generate synthetic travel-time data from this surrogate model and attempt to reconstruct the underlying velocity structure using the rj-MCMC algorithm. The method successfully recovers most the main features of the anomalies when compared to the known surrogate model, despite some issues with overly-determined regions. This demonstrates its ability to adapt model complexity and resolve spatial variations from limited data.

In addition, we apply the same methodology to real seismic data from Denmark. While no ground truth model is available for validation, the results are internally consistent and reflect the spatial variability expected from the data coverage.

Overall, this study demonstrates that rj-MCMC with Voronoi parameterization provides a flexible and robust framework for transdimensional inversion of seismic data, particularly in settings with uneven data constraints.

I. INTRODUCTION

Mapping the underground is important for many reasons, including assessing the risk of natural hazards such as earthquakes, exploring resources like oil and groundwater, planning infrastructure such as buildings and tunnels, and understanding ice sheet dynamics [1]. However, it is not an easily accessible task, as the subsurface cannot be observed directly. Instead, geophysicists use seismic tomography, where networks of seismometer stations provide information about wave velocities within parts of the Earth's interior [2]. However, the data are inherently limited, as measurements are only obtained along the paths between seismic stations. This means that the subsurface is not sampled continuously, but only along discrete ray paths connecting sources and receivers. Consequently, large regions remain poorly constrained, and the available data must be used inversely to infer a complete velocity field of the subsurface. Since different materials affect seismic wave speeds differently, reconstructing this velocity field is essential for mapping underground structures. Bayesian inference provides a natural framework for addressing such inverse problems, as it allows uncertainties to be quantified and incorporates prior knowledge about the system. Rather than identifying a single best-fit model, Markov Chain Monte Carlo (MCMC) methods enable sampling from the full posterior distribution of possible models. The reversible jump MCMC (rj-MCMC) algorithm extends this approach by allowing the number of model parameters (e.g. cells) to vary, thereby enabling automatic control of model complexity. A flexible way to represent spatial structure is through Voronoi cells, which partition the velocity field into regions with constant velocity. In this project, we first apply an rj-MCMC method to a surrogate data set which is a simplified model of an underground consisting of three zones with perturbed slowness. We then use this surrogate model to generate synthetic seismometer data using a forward model. This seismometer data is then used to reconstruct the subsurface structure with the three perturbed zones back using the rj-MCMC method to test the performance of the rj-MCMC method. As an addition

we also attempt using the method on real seismic data from Denmark's underground and map which materials it consist of.

II. METHODS

A. Discretization and Voronoi cells

The beauty of the rj-MCMC method lies in the spatially variant discretization. This process is done by using Voronoi Cells which are defined as a polygon around a nuclei that encloses the area in which that nuclei is the closest of all nucleus. This allows for the grid to be spanned by polygons of varying shapes and sizes.

B. Forward Problem

To estimate seismic travel times through the subsurface of Denmark, we assume straight rays throughout the cells. This is a valid simplification when velocity variations are moderate, as strong velocity contrasts that would significantly bend ray paths are not expected, and also quite complex to take account for. The subsurface velocity field is parametrized using Voronoi cells [3], where each cell has a constant slowness value $s_i = 1/v_i$. Given a set of n nuclei $\{\mathbf{c}_i\}$ and associated slowness values $\{s_i\}$, the predicted travel time of the j -th ray is computed by sampling the ray path at evenly spaced intervals and summing the slowness contributions [2]:

$$t_j = \delta l_j \sum_{k \in \text{ray}_j} s(R(\mathbf{x}_k)) \quad (1)$$

where $\mathbf{x}_k$ are the sample point coordinates along the ray, $R(\mathbf{x}_k)$ is the index of the Voronoi cell (the nearest nucleus) containing that point, and δl_j is the fixed sub-segment length. In practice, each ray is sampled at approximately 5 points per km, with all sample points for all rays precomputed and stored prior to the inversion. Cell assignments are obtained through a k-d tree nearest-neighbour method

. To stabilise the inversion near the domain boundaries, a fixed ring of ghost nuclei is placed just outside the model domain. These are assigned the background slowness s_0 and are never perturbed during the inversion. They prevent boundary Voronoi cells from growing unconstrained into unsampled regions outside the study area.

C. Bayesian Formulation

The inversion is formulated within a Bayesian framework, where the goal is to explore the posterior probability distribution of the model parameters given the observed data [4?]. Using Bayes' theorem, the posterior is proportional to the product of the likelihood and the prior:

$$p(\mathbf{m} \mid \mathbf{d}_{\text{obs}}) \propto p(\mathbf{d}_{\text{obs}} \mid \mathbf{m}) p(\mathbf{m}) \quad (2)$$

where $\mathbf{m} = (\mathbf{c}, \mathbf{s})$ is the model vector containing the nucleus positions and slowness values, and $\mathbf{d}_{\text{obs}}$ is the vector of observed travel times. Because the number of Voronoi cells n is itself treated as an unknown, the posterior is defined across model spaces of varying dimension [2].

1. Likelihood

The likelihood function quantifies how well a given model reproduces the observed travel times. Assuming Gaussian data noise with standard deviation σ_d , the likelihood takes the form:

$$p(\mathbf{d}_{\text{obs}} \mid \mathbf{m}) \propto \exp\left(-\frac{\phi(\mathbf{m})}{2}\right) \quad (3)$$

$$\phi(\mathbf{m}) = \sum_{j=1}^{N_{\text{rays}}} \left(\frac{t_j(\mathbf{m}) - d_j^{\text{obs}}}{\sigma_d} \right)^2 \quad (4)$$

where $t_j(\mathbf{m})$ is the predicted travel time of the j -th ray given by equation 1, and σ_d is the estimated standard deviation of the data noise.

2. Prior

We use uniform prior distributions for all model parameters [2]. The number of cells n is drawn from a uniform distribution over the interval $[N_{\text{min}}, N_{\text{max}}]$. Given n , the slowness of each cell is drawn independently and uniformly from the interval $[s_{\text{min}}, s_{\text{max}}]$, and nucleus positions are drawn uniformly over the model domain. The full prior is:

$$p(\mathbf{m}) \propto \frac{1}{\Delta s^n \Delta n} \quad (5)$$

where $\Delta s = s_{\text{max}} - s_{\text{min}}$ and $\Delta n = N_{\text{max}} - N_{\text{min}}$. In our implementation, the slowness is bounded between $s_{\text{min}} = 0.18$ s/km and $s_{\text{max}} = 0.62$ s/km, corresponding to velocities between approximately 1.6 and 5.6 km/s, and the number of cells is allowed to vary between $N_{\text{min}} = 2$ and $N_{\text{max}} = 200$.

D. The rj-MCMC Sampler

To sample the transdimensional posterior distribution, we use the reversible jump Markov chain Monte Carlo algorithm of Green [5], which extends the standard Metropolis-Hastings algorithm [6, 7] to allow transitions between model spaces of different dimension. Following Bodin and Sambridge [2], the chain alternates between two types of steps:

- **Slowness update** (every even step) the slowness of a randomly chosen cell is perturbed by drawing $\delta s \sim \mathcal{N}(0, \sigma_s^2)$ and setting $s'_k = s_k + \delta s$. The Voronoi cell configuration is unchanged.
- **Geometry move** (every odd step) one of three moves is chosen with equal probability:
 - **Birth** a new nucleus is placed at a position drawn uniformly over the domain. Its slowness is set to $s'_{n+1} = s_i + u_v$, where s_i is the slowness at the birth location in the current configuration and $u_v \sim \mathcal{N}(0, \sigma_2^2)$.
 - **Death** a randomly chosen nucleus is removed. The surrounding cells expand to fill the gap.
 - **Nucleus move** the position of a randomly chosen nucleus is perturbed by drawing from $\mathcal{N}(\mathbf{0}, \sigma_c^2 \mathbf{I})$. Its slowness travels with it.

For the fixed-dimension moves the proposal distributions are symmetric and the acceptance probability reduces to the likelihood ratio [2]:

$$\alpha(\mathbf{m}' \mid \mathbf{m}) = \min\left(1, \frac{p(\mathbf{d}_{\text{obs}} \mid \mathbf{m}')}{p(\mathbf{d}_{\text{obs}} \mid \mathbf{m})}\right) \quad (6)$$

For the transdimensional moves the proposals are asymmetric and the acceptance probability must account for the change in dimension. The log acceptance probability for a birth step is [2]:

$$\log \alpha_{\text{birth}} = \log \frac{\sigma_2 \sqrt{2\pi}}{\Delta s} + \frac{u_v^2}{2\sigma_2^2} + \log \frac{p(\mathbf{d}_{\text{obs}} \mid \mathbf{m}')}{p(\mathbf{d}_{\text{obs}} \mid \mathbf{m})} \quad (7)$$

where u_v is the Gaussian deviate used to generate the new cell's slowness and σ_2 is the corresponding proposal standard deviation. The death acceptance probability is the inverse of the birth. This naturally discourages adding unnecessary cells the algorithm always prefers one large cell over two small cells with similar slowness values[8]. To improve efficiency, we apply a two stage delayed rejection scheme [9, 10] to the fixed-dimension moves. If a proposed move is rejected, a second move is attempted with a smaller step size.

E. Surrogate data generation

To test the method we created a surrogate dataset. This was done by creating a grid of size $N \times N$ with constant values for slowness in all grid points. Then defining three zones with perturbed slowness (a triangle, a circle and a square). The ray coverage is made up of 48 "sensors" evenly distributed along the edges of the grid acting as both senders and receivers. The forward model was then applied to this grid and a Gaussian noise was added to obtain a surrogate dataset.

III. RESULTS

A. Surrogate data and model

To test the correctness of our implementation of the rj-mcmc algorithm we performed simulation on the surrogate dataset we set up. This test was done by running a single chain with 50k steps and discarding the first 20k as burn-in. Samples were then taken ever 10 steps. The plot here is the final sample.

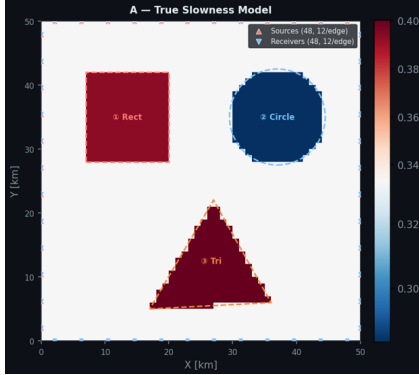


FIG. 1. Plot of the true model used to get the surrogate dataset

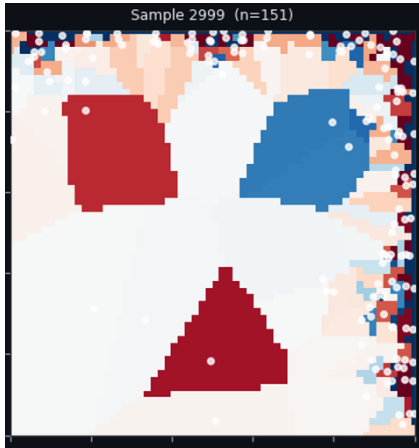


FIG. 2. Voronoi cell gallery after 3000 samples (50k steps)

B. Denmark model

The Denmark dataset was constructed from vertical-component waveform data from the Danish Seismological Network (DK), obtained through the EIDA FDSN web services [11, 12]. Interstation travel times were extracted from the processed waveforms and used as input for the inversion.

Two RJMCMC ensembles were generated. The first consisted of 20 independent chains with 100,000 steps each, and the second consisted of 11 independent chains with 200,000 steps each. For the 100,000-step ensemble, the first 20,000 steps were discarded as burn-in and every 20th sample was kept. For the 200,000-step ensemble, the first 40,000 steps were discarded and every 40th sample was kept. The retained samples were combined to estimate posterior mean velocity and posterior uncertainty.

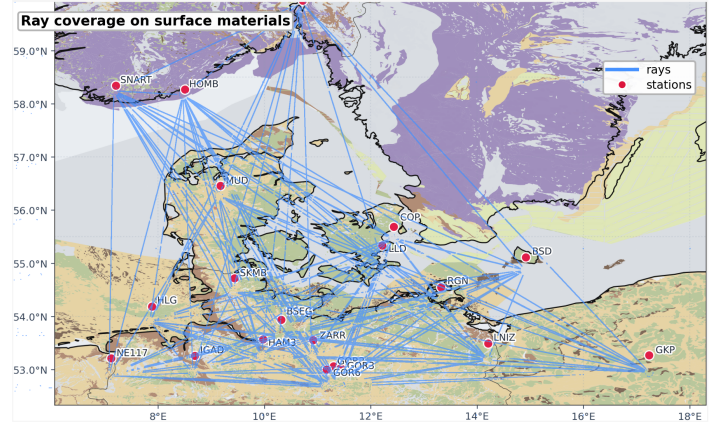


FIG. 3. Ray coverage shown on top of the harmonized surface-material classes used as geological context for the inversion.

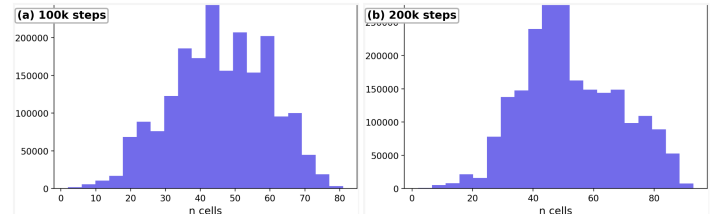


FIG. 4. Posterior distributions for the number of Voronoi cells for the 100,000-step and 200,000-step ensembles.

Figure 3 shows the station geometry and ray coverage together with the harmonized surface-material map used as geological context. Figure 4 shows the posterior distributions for the number of Voronoi cells for the two ensembles. The posterior mean velocity and posterior uncertainty for the 100,000-step and 200,000-step ensembles are shown in Figures 5 and 6. The corresponding maps for each ensemble are shown separately in Appendix A.

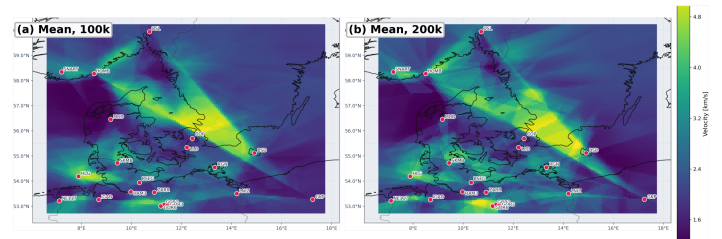


FIG. 5. Posterior mean velocity for the 100,000-step and 200,000-step ensembles shown with a shared color scale.

IV. DISCUSSION

A. Surrogate model

The results from the surrogate simulation shows clear resemblance of the expected result and qualitatively confirms that our model is able to recover some of the vital information from a surrogate dataset. The results are, however, not flawless. It is clear to see that there seems to be a lapse in precision along the left and top boundary of the grid.

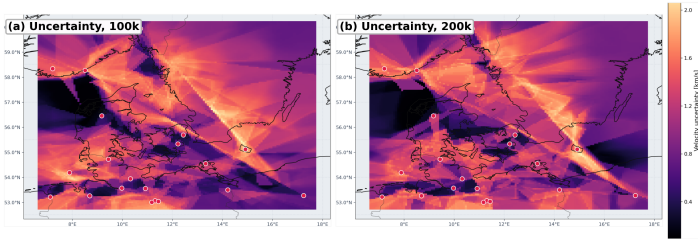


FIG. 6. Posterior velocity uncertainty for the 100,000-step and 200,000-step ensembles.

We believe this issue was caused by the way we set up the sender/receivers along the edges of the grid. This combined with the fact that the blue colored anomaly, with a negatively perturbed slowness, inhabits the top left corner and both other anomalies have a positively perturbed slowness makes for a situation where the edges are overly determined and thus requires higher resolution. This mistake meant the dimensionality of our model went higher than expected, as high as 150 cells, making it computationally costly. We believe that this issue could be fixed and results even more consistent with the surrogate model and more computationally efficient could be achieved. In such a scenario the simulations could be performed across a high number of chains and an average solution grid would allow for uncertainties on results and a way to measure correctness quantitatively.

Despite this issue the results still showcase the algorithms ability to describe a mixed-determined problem and discretize in a way that adapts to the problem at hand. All this being done through the addition of a single parameter, the number of cells, is, in our opinion, quite remarkable.

B. Mapping the Danish underground

When analysing the results from our attempt to map the Danish underground using *rj-MCMC*, we were not able to find enough directly comparable reference models. Because of this, we cannot properly evaluate the geological correctness of the recovered structures, either qualitatively or quantitatively. What we can evaluate instead is whether the method behaves consistently, whether the chains move towards similar results, and whether the model reflects the mixed-determined nature of the data.

A central challenge in this problem is that the data coverage is highly uneven across the study area. Some regions are crossed by many rays, while others are only weakly sampled. This means the inversion is not equally well constrained everywhere, but is instead a mixed-determined problem. In that setting, one of the main advantages of the *rj-MCMC* Voronoi method is its spatially variant discretization. Since the model is built from Voronoi cells whose number and geometry can change during sampling, the discretization can adapt to the information content of the data. In better determined areas, the method can place more and smaller cells and therefore resolve more detail, while in underdetermined areas it can keep a simpler description and avoid forcing unnecessary structure.

This is also reflected in the comparison between the 100k and 200k ensembles. The 200k run has had more time to explore the model space, and its posterior on the number

of Voronoi cells extends further towards higher cell counts than the 100k run. This suggests that the shorter run has not explored the full range of model complexity to the same degree. At the same time, the posterior mean velocity maps from the two runs remain broadly similar, which indicates that the large-scale features are relatively stable. The 200k result mainly appears more nuanced, rather than completely different.

The uncertainty maps show the same pattern. Areas with stronger ray coverage are better constrained, while areas with fewer rays remain more uncertain. This is exactly where the spatially variant discretization becomes important. A fixed discretization would spend the same model resolution everywhere, including in regions where the data do not justify it. In contrast, the Voronoi approach allows the inversion to focus its complexity where the problem is better determined, which makes it especially well suited for mixed-determined inverse problems such as this one.

Overall, the Denmark results suggest that the method is internally consistent and that the main large-scale structures are reproducible between runs. At the same time, the comparison between 100k and 200k steps shows that further exploration of the posterior would likely still be useful, although that comes at a significant computational cost. The results therefore show both the strength of the method and the limitations set by data coverage and available computation time.

V. CONCLUSION

In this project, we have applied reversible jump Markov Chain Monte Carlo with a Voronoi cell parameterization to the transdimensional inversion of seismic travel-time data. Validation on a synthetic surrogate dataset confirmed that the algorithm successfully recovers the main structural features of an unknown subsurface, despite some over-determination near the grid boundaries caused by the sensor placement. This drove model complexity higher than necessary, though the core performance of the method remained clear. When applied to real data from the Danish Seismological Network, both the 100,000-step and 200,000-step ensembles produced broadly consistent large-scale velocity structures, with uncertainty appropriately concentrated in poorly sampled regions. The longer run yielded a more nuanced posterior, suggesting that further exploration of model space would be beneficial, albeit at significant computational cost. Overall, *rj-MCMC* with Voronoi parameterization proves to be a flexible and robust framework for seismic inversion, particularly in mixed-determined settings with uneven data coverage. Its ability to adaptively allocate model complexity where the data demand it makes it a compelling approach for future geophysical imaging studies.

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Appendix A: Additional figures

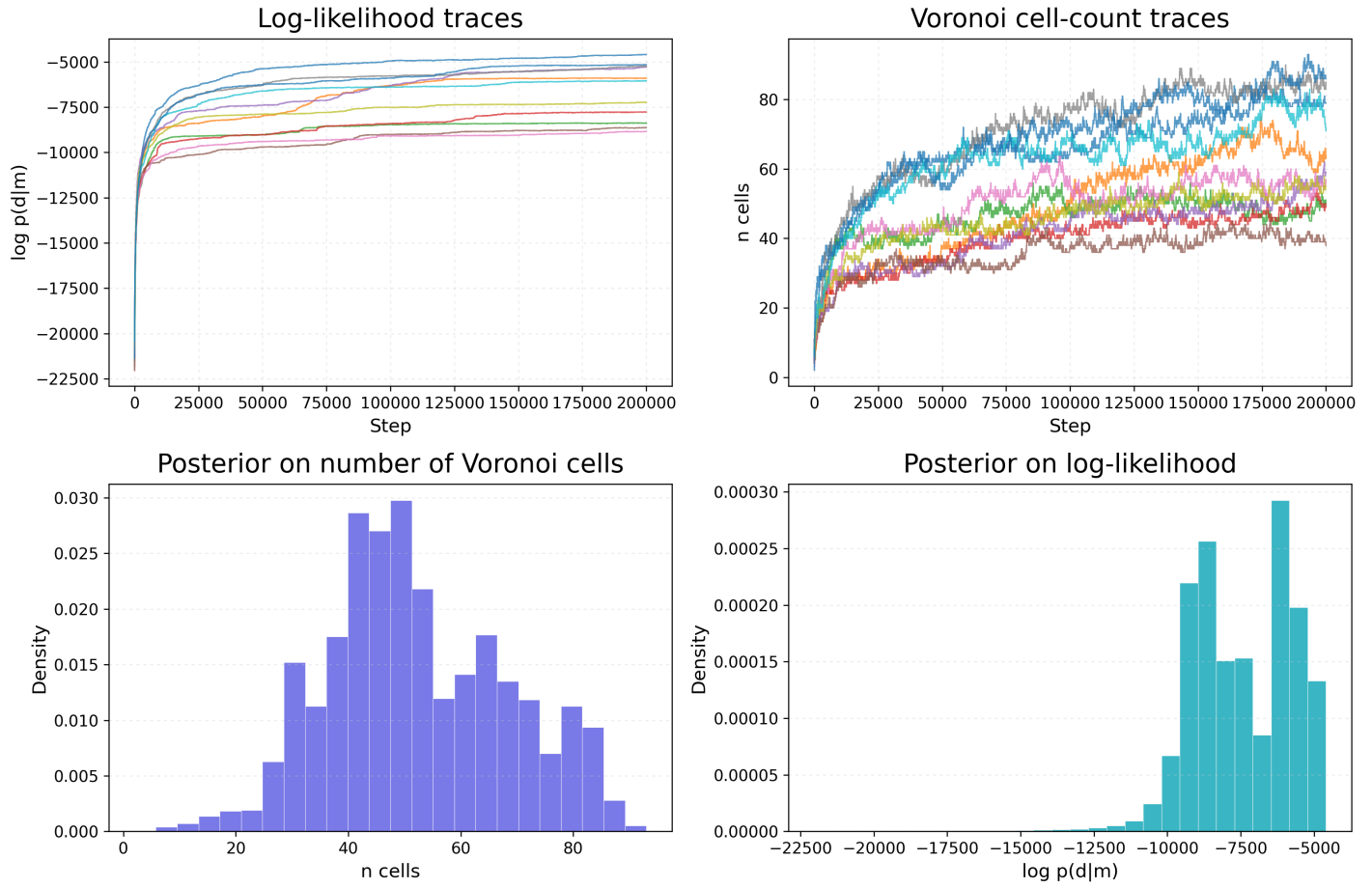


FIG. 7. Diagnostic plots for the 100,000-step ensemble. The panels show the log-likelihood traces, Voronoi cell-count traces, posterior distribution of the number of Voronoi cells, and posterior distribution of the log-likelihood values.

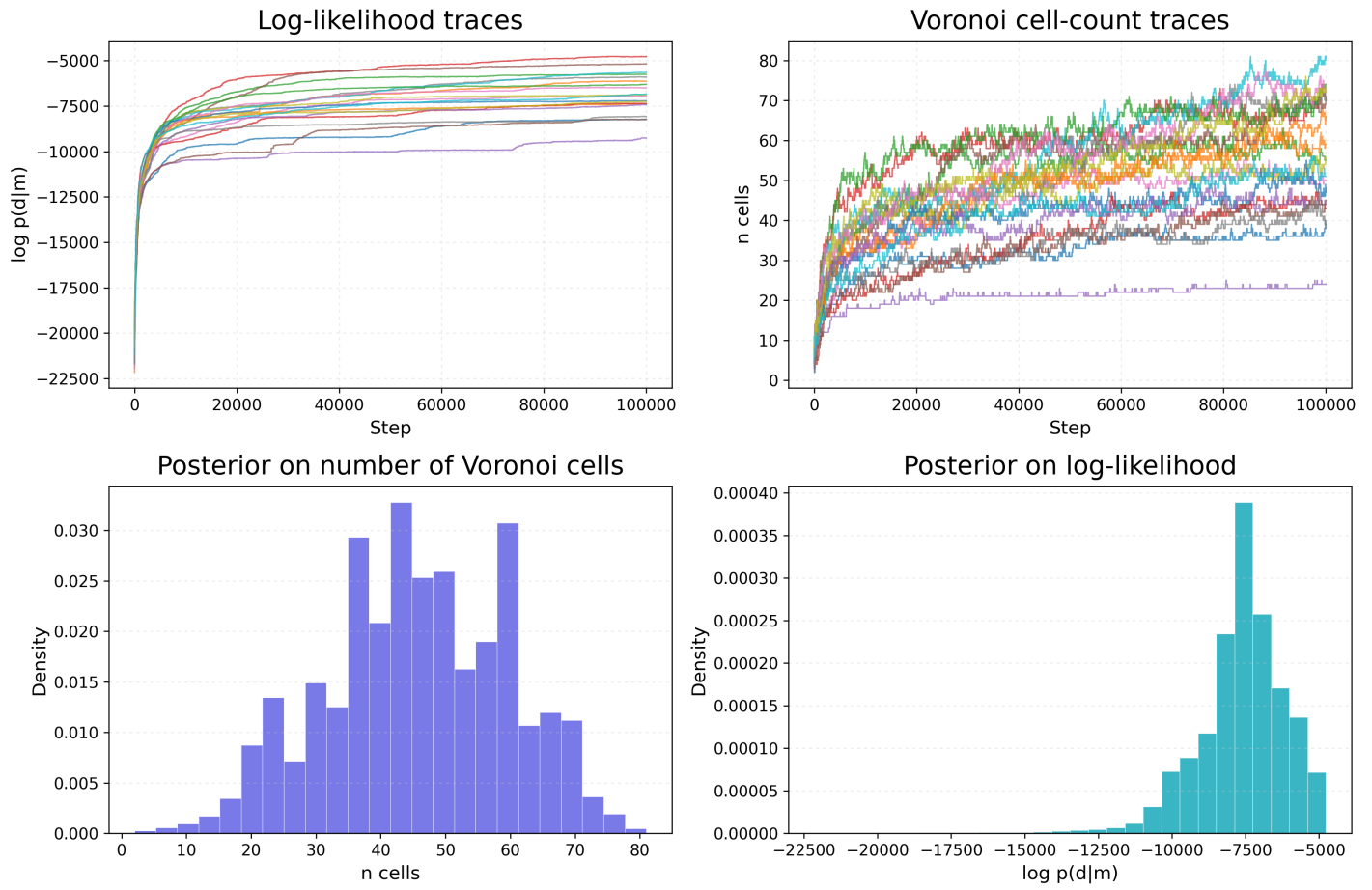


FIG. 8. Diagnostic plots for the 200,000-step ensemble. The panels show the log-likelihood traces, Voronoi cell-count traces, posterior distribution of the number of Voronoi cells, and posterior distribution of the log-likelihood values.

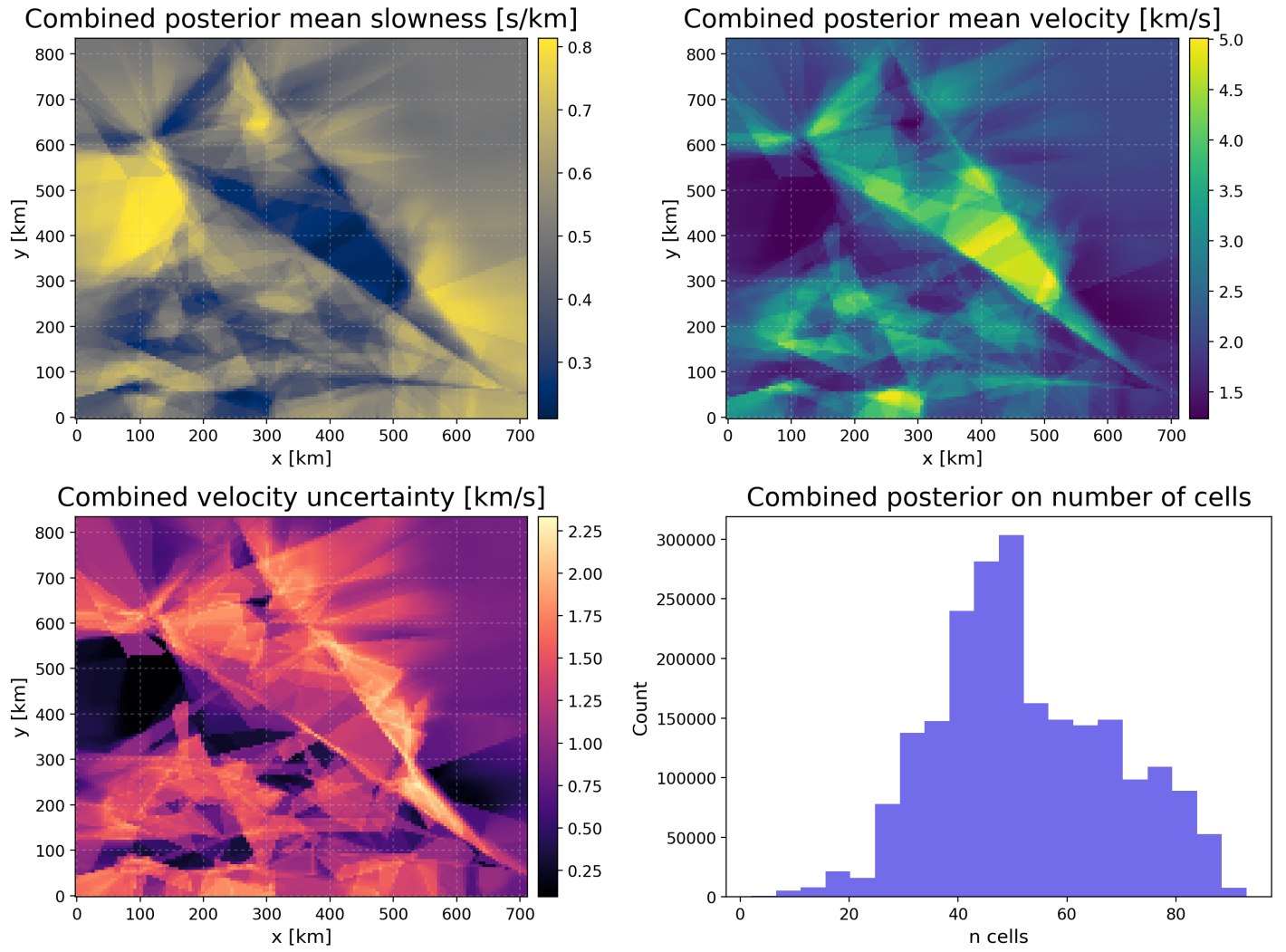


FIG. 9. Posterior summary for the 100,000-step ensemble, including posterior mean slowness, posterior mean velocity, posterior velocity uncertainty, and posterior distribution of the number of Voronoi cells.

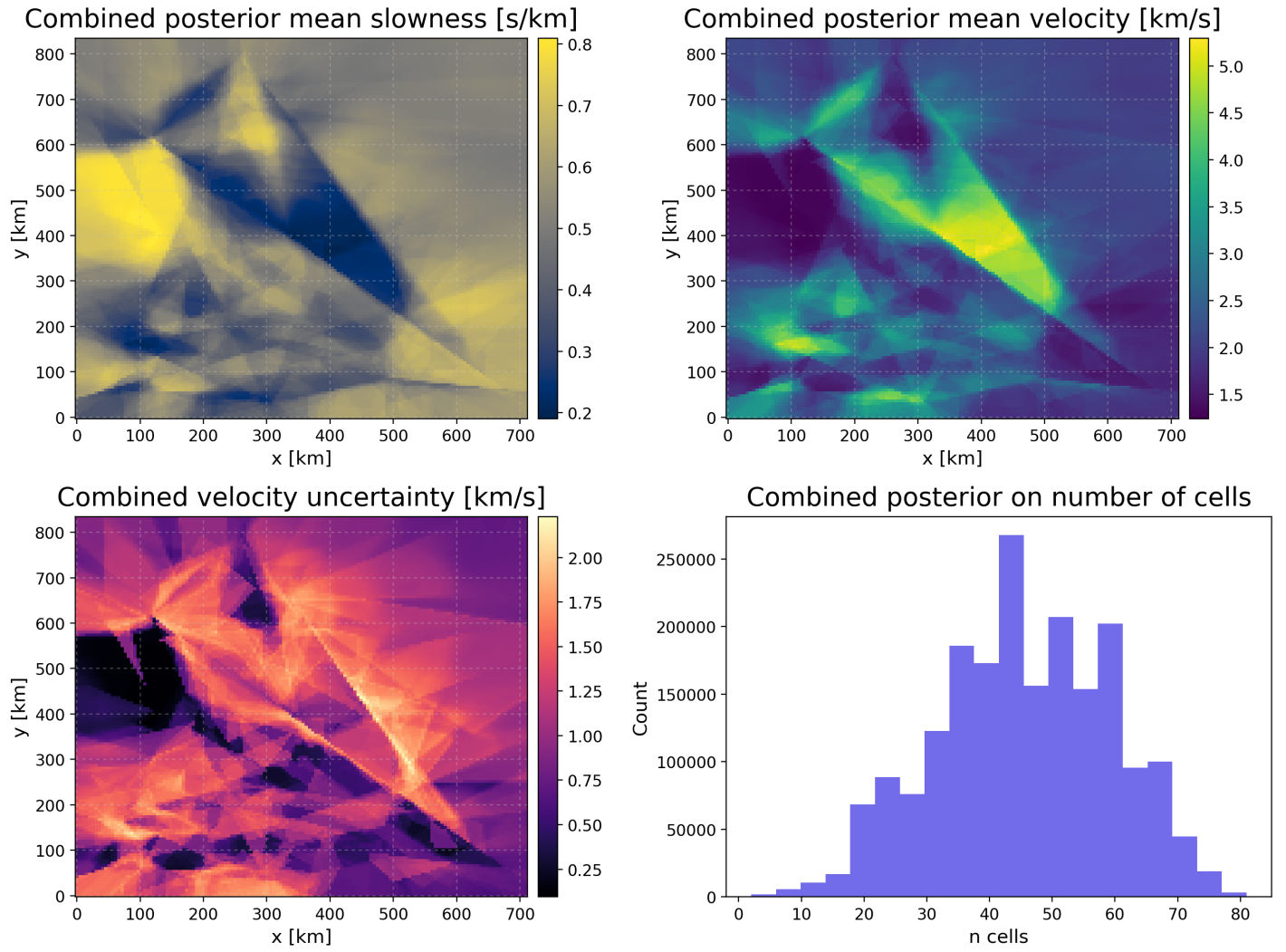


FIG. 10. Posterior summary for the 200,000-step ensemble, including posterior mean slowness, posterior mean velocity, posterior velocity uncertainty, and posterior distribution of the number of Voronoi cells.